

ENCODE

Scientific foundation v0

An information-to-cognition system, evaluated through independent performance

Research and design study
Prepared for Advay
3 October 2026

The north star

Minimum total time to durable understanding, accurate recall, procedural competence, and transfer.

Scope

A critical evidence review with 62 academic and curriculum references, proposed teaching and language specifications, and a practical experimental program. No app or repository changes. No claim that the protocol has already been validated on the learner.

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Reading note: wide tables from the editable Markdown are rendered as labeled records here to preserve readable type. All fields and decisions are retained.

Review method and evidence conventions

This is a targeted, critical evidence review, not a registered systematic review or a new meta-analysis. Searches covered cognitive and educational psychology, mathematics education, computing education, learner modeling, and controlled AI-tutoring research. Search phrases combined each requested technique with review, meta-analysis, experiment, transfer, retention, novice, and boundary conditions. Searches and citation following were conducted on 3 October 2026. Publisher pages, author manuscripts, academic repositories, PubMed/ERIC records, and official IB materials were preferred. Recent revisions were sought alongside established work. No claim of exhaustive coverage, independent duplicate screening, or a PRISMA search flow is made.

The bibliography records the level of source inspection. **F** means relevant full-text sections were inspected, not necessarily every page; **A** means an abstract or academic record was inspected; **O** means an official curriculum document or brief. Abstract-only inspection limits methodological appraisal. The source register makes these limitations explicit. No neuroscience result is needed to justify the proposed product.

Evidence grades apply to a particular claim and outcome, not to a technique in all circumstances:

- **STRONG:** convergent reviews and controlled evidence for the stated outcome; still subject to boundaries.
- **MODERATE:** credible synthesis or controlled evidence with narrower coverage, heterogeneity, or important implementation uncertainty.
- **CONTEXT-DEPENDENT:** the direction or magnitude depends materially on knowledge, task, comparator, timing, or implementation.
- **HYPOTHESIS:** a plausible ENCODE design choice or learner-specific prediction requiring testing.

Effect sizes are standardized contrasts within different studies. They are not percentages of learning, cannot be added, and should not be used to rank techniques across unrelated tasks. Meta-analytic moderators are generally associations between studies, not isolated randomized tests of a component. Most evidence concerns groups, not Advay. The referenced conversation supplied a project brief, not scored learning trials. Its assistant-generated learner interpretations are not independent evidence.

1. Executive scientific model

ENCODE should optimize **time to demonstrated, durable, independent competence**, rather than explanation quality, answer familiarity, lesson completion, or enjoyment alone. The reliable foundation is ordinary instructional engineering: accurate material, suitable examples, effortful practice, informative correction, practice distributed over time, and assessment that removes help. The broad review by Dunlosky and colleagues supports practice testing and distributed practice while giving more qualified support to elaborative interrogation, self-explanation, and interleaving. [R01](#)

The compiler metaphor is useful at the system boundary: validated source material enters; a sequence of learning opportunities and assessments leaves. It does not mean the tutor directly writes representations into a brain. A learner constructs and revises representations through activity. Attention, comprehension, retrieval, practice history, and task demands constrain that process.

An engineering objective is: minimize total active instruction, practice, correction, and review time, subject to explicit performance requirements at specified delays and in specified task families. Maintain a separate vector of requirements for recall, explanation, discrimination, procedure, application, transfer, and speed. If time and transfer conflict, report that tradeoff. A scalar engagement score cannot substitute for these constraints.

Three findings that constrain the design

Retrieval is valuable, but its comparator matters. A classroom meta-analysis synthesized 222 studies and 48,478 learners, finding an average testing benefit of $g = 0.499$ across heterogeneous comparisons. That does not establish superiority over every form of active instruction. A 2025 synthesis comparing retrieval with elaborative encoding found a smaller overall advantage, $g = 0.14$, across 44 studies; feedback and the elaborative comparator mattered. ENCODE should combine useful explanation with retrieval, not replace understanding with quizzes. [R02](#), [R04](#)

Domain evidence can revise the general rule. A 2025 mathematics synthesis found spacing benefits of $g = 0.28$ across 27 studies. Its retrieval-versus-restudy estimate was $g = 0.18$ across only seven studies, with an interval crossing zero. This is inconclusive evidence for that specific mathematics contrast, not proof that mathematical retrieval is useless. [R07](#)

Assisted performance can conceal dependence. In a high-school mathematics trial, a relatively unrestricted GPT interface improved assisted practice yet worsened subsequent unassisted performance relative to control. A guarded tutor avoided that decrement; its unassisted performance was statistically indistinguishable from control. This establishes a material failure risk, not a guarantee that guardrails alone improve learning. [R54](#)

Scientific translation of the metaphor

Product phrase	Legitimate interpretation	What ENCODE may infer
Encode information	Attend to and process material so later performance becomes possible	Whether fresh responses improve; not neural encoding efficiency
Build a schema	Organize relations, conditions, and procedures so they support explanation and action	Convergent explanation, discrimination, and application evidence
Strengthen retrieval	Increase accessibility under particular cues and delays	Probability of successful independent recall under stated conditions
Increase storage strength	A theoretical distinction from current accessibility	Long-delay observations; no directly measured storage-strength meter
Reconsolidate	A proposed memory-restabilization mechanism under particular conditions	No direct inference from a corrected answer
Automatize	Improve accurate execution with less deliberation on practiced components	Correct-response latency and interference/error patterns
Program transfer	Enable use beyond trained items	Performance on explicitly novel, independently scored tasks

Learning-versus-performance and metacognitive reviews support separating temporary accessibility from durable learning. Human reconsolidation evidence does not license labeling every retrieval-and-feedback episode a reconsolidation event. [R31](#), [R32](#), [R42](#)

2. Personalized learner profile: observations and hypotheses

The supplied observations are requirements and self-reported impressions, not established cognitive traits. The installed personal model was consulted for task-relevant judgment and evidence discipline. Its provisional learning hypotheses do not replace direct baseline measurement. No neurological, medical, personality-type, or learning-style explanation is warranted. Preference matching is not an established route to better learning. R43

Concise, direct explanations are strongly preferred

Plausible interpretation: Filler may impose avoidable reading time; excessive compression may omit dependencies

Evidence-backed implication: Remove irrelevant material; preserve necessary relations and examples

Uncertainty and direct test: Preference is credible; learning benefit is unmeasured. Compare equal-content concise and fuller versions using delayed explanation and transfer

Causal models and concrete examples appear helpful

Plausible interpretation: Relations may support organized understanding; examples may anchor unfamiliar abstractions

Evidence-backed implication: Map the example explicitly to the rule; test conditions and counterexamples

Uncertainty and direct test: No scored comparison. Compare causal-plus-example with a competent definition-and-example lesson

One question at a time is preferred

Plausible interpretation: Smaller response units may reduce navigation and competing demands

Evidence-backed implication: Use one main response request per turn, while keeping context visible

Uncertainty and direct test: Could slow practice or fragment multi-step reasoning. Measure total time, coherence, and delayed performance

Explanation, example, tracing, modification, then construction appears better for beginner coding

Plausible interpretation: Simultaneous syntax, logic, and tooling demands may exceed current prior knowledge

Evidence-backed implication: Start with an explicit execution model and fade support toward independent code

Uncertainty and direct test: Sequence is not proven for Advay. Compare matched tiny-program lessons and unassisted generation

Independent competence matters more than familiarity

Plausible interpretation: The intended outcome includes performance without answer support

Evidence-backed implication: Remove tutor assistance during checks; record first attempts and hint dependence

Uncertainty and direct test: A goal, not a demonstrated skill. Measure fresh tasks and delayed recall

Many projects and tools are pursued simultaneously

Plausible interpretation: Switching and choice overhead may consume scarce study time

Evidence-backed implication: Keep a stable learning environment and a small next-action queue

Uncertainty and direct test: No evidence of an attention disorder or fixed capacity. Compare continuity versus normal workflow using interruptions and effective study time

Conceptual/business reasoning seems stronger than foundational math/programming fluency

Plausible interpretation: Unequal experience or practice may explain differences

Evidence-backed implication: Diagnose prerequisites separately by domain and dimension

Uncertainty and direct test: Comparative ranking is unverified. Obtain independent baselines before allocating difficulty or labeling strengths

Personalization should first change **presentation and workflow**, then change instructional policy only when measured outcomes justify it. Respect concise wording and one-turn questions unless they remove essential explanation or coherent task context. Do not infer that a preference for easy interaction means a preference against effortful retrieval. Do not infer a permanent learner type from a temporary skill gap.

3. Evidence map

The mechanism column gives a plausible account, not proof that a single mechanism explains an effect. The final column is an inference for ENCODE. Repeated citations refer to the same evidence base, not independent replications.

Retrieval practice R02, R03, R04

Proposed mechanism: Reconstruct information and strengthen accessible routes; reveal gaps

Demonstrated outcome: Better retention across many classroom comparisons

Grade for stated claim: STRONG broadly; comparator-sensitive

Boundary conditions: Feedback, format, delay, material, and quality of alternative instruction matter

ENCODE implication: Retrieve without notes; correct errors; retain useful elaboration

Spacing R05, R06, R07

Proposed mechanism: Reactivation after accessibility changes; multiple learning occasions

Demonstrated outcome: Better delayed performance than massing

Grade for stated claim: STRONG; MODERATE domain precision

Boundary conditions: Target retention interval and content matter; no universal interval

ENCODE implication: Spread practice across days; fit intervals to actual delayed success

Successive relearning R08, R09

Proposed mechanism: Successful retrieval re-established across spaced sessions

Demonstrated outcome: Course-test and long-term retention benefits

Grade for stated claim: MODERATE

Boundary conditions: Most direct evidence concerns learnable factual/conceptual answers; time costs vary

ENCODE implication: Use recurring recall-to-criterion for important knowledge, not one-session repetition alone

Interleaving R10, R11

Proposed mechanism: Contrast categories and practice choosing the right method

Demonstrated outcome: Benefits for some category and mathematics tasks

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Similarity and task class matter; blocked practice can win for words

ENCODE implication: Establish basic execution, then mix confusable methods without announcing which to use

Generation R12

Proposed mechanism: Active production rather than reading

Demonstrated outcome: Memory benefit across generation tasks

Grade for stated claim: MODERATE

Boundary conditions: Producing a word/answer is not equivalent to generating a whole novice program

ENCODE implication: Ask for small predictions or missing steps when prerequisites exist

Pretesting R13, R14

Proposed mechanism: Orient attention and create questions answered by later study

Demonstrated outcome: Benefits often concentrated on prequestioned material

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: A meta-analysis found specific $g = 0.54$ but general $g = 0.04$; errors require correction

ENCODE implication: Optional short diagnostic guess; do not claim broad transfer or prolong guessing

Worked examples R15, R16

Proposed mechanism: Reduce unproductive search while exposing solution structure

Demonstrated outcome: Mathematics performance benefit; $g = 0.48$ in a 2023 synthesis

Grade for stated claim: STRONG for novice example support

Boundary conditions: Prior knowledge, example quality, comparator, and task complexity matter

ENCODE implication: Demonstrate a correct solution with its conditions before unsupported complex construction

Example fading and completion R17, R18

Proposed mechanism: Transfer responsibility for steps gradually

Demonstrated outcome: Controlled evidence for transition toward problem solving

Grade for stated claim: MODERATE

Boundary conditions: Near-transfer evidence is firmer than universal far transfer

ENCODE implication: Remove steps when fresh independent work supports doing so

Element interactivity R16, R62

Proposed mechanism: Dependencies must be coordinated relative to available schemas

Demonstrated outcome: Explains important instructional reversals and load effects

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Interactivity is learner-relative; not a fixed count of words

ENCODE implication: Diagnose prerequisites; externalize linked quantities/state; avoid numeric load claims

Chunking and schemas R16, R19

Proposed mechanism: Organize familiar components into functional units

Demonstrated outcome: Established theoretical and experimental support for knowledge-dependent processing

Grade for stated claim: MODERATE as design rationale

Boundary conditions: No fixed four-item instructional limit; chunk meaning depends on knowledge

ENCODE implication: Teach meaningful relations and subgoals, not arbitrary tiny fragments

Self-explanation R20, R15, R18

Proposed mechanism: Connect steps to principles and detect gaps

Demonstrated outcome: Broad positive synthesis; mathematics moderator raises limits

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Extra prompts are not always beneficial; unsupported explanations can be wrong

ENCODE implication: Use a targeted why/condition question at a bottleneck, not every line

Elaborative interrogation R01

Proposed mechanism: Link a fact to reasons or existing knowledge

Demonstrated outcome: Qualified benefits for suitable factual materials

Grade for stated claim: MODERATE

Boundary conditions: Requires relevant knowledge; invented explanations are dangerous

ENCODE implication: Ask why only where an explanation can be checked

Concrete examples and fading R21

Proposed mechanism: Anchor meaning, then map toward abstraction

Demonstrated outcome: Promising mathematics/science review evidence

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Extraneous features may dominate; not always concrete-first

ENCODE implication: Identify invariant features and test an abstract/changed-context case

Analogy and contrasting cases R22

Proposed mechanism: Compare structure and discriminating features

Demonstrated outcome: Case-comparison synthesis supports learning benefits

Grade for stated claim: MODERATE

Boundary conditions: Mapping must be explicit; irrelevant similarity can mislead

ENCODE implication: Show matched positive/negative cases and ask what changes the rule

Dual representation and multimedia R23

Proposed mechanism: Coordinate complementary verbal and visual information

Demonstrated outcome: Review synthesis supports selected multimedia principles

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Redundant or decorative media can burden learning

ENCODE implication: Use diagrams/state tables where they explain relations; avoid learning-style claims

Signaling R25

Proposed mechanism: Direct attention to relevant structure

Demonstrated outcome: Positive multimedia meta-analysis

Grade for stated claim: MODERATE

Boundary conditions: Highlighting everything removes the signal

ENCODE implication: Highlight the changed variable, subgoal, or causal link

Segmenting R24

Proposed mechanism: Permit processing and learner control over portions

Demonstrated outcome: Retention/transfer gains in multimedia studies

Grade for stated claim: MODERATE

Boundary conditions: Often increases study time; segmenting may destroy continuity

ENCODE implication: Pause at conceptual boundaries; compare benefits per minute

Modality and transient information R26

Proposed mechanism: Spoken information disappears unless retained/externalized

Demonstrated outcome: Long spoken instructions can lose advantages over persistent text

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Duration, visual demands, pace, and prior knowledge matter

ENCODE implication: Keep equations, diagrams, and code visible; let speech pause/replay

Informative feedback R27, R28

Proposed mechanism: Correct discrepancies and supply actionable next steps

Demonstrated outcome: Average benefit with substantial heterogeneity

Grade for stated claim: STRONG for useful corrective information

Boundary conditions: Timing and specificity vary; praise is not equivalent to task correction

ENCODE implication: Explain the error and next step; test a fresh response after repair

Feedback timing R28, R34

Proposed mechanism: Balance correction with an opportunity to retrieve/diagnose

Demonstrated outcome: No universal best delay

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Novice misunderstanding differs from consolidated recall

ENCODE implication: Correct unsupported misconceptions promptly; delay the reveal briefly for a feasible attempt

Mastery learning R29

Proposed mechanism: Correct gaps before dependent material; allocate additional practice

Demonstrated outcome: Controlled program evidence supports achievement gains

Grade for stated claim: MODERATE

Boundary conditions: Extra time and completion/pacing costs; criterion definition matters

ENCODE implication: Gate prerequisites with relevant evidence, not a global score

Adaptive difficulty R30

Proposed mechanism: Match demand to current ability and useful errors

Demonstrated outcome: An 85% optimum exists in particular computational classification models

Grade for stated claim: HYPOTHESIS for tutor thresholds

Boundary conditions: Not a general human-teaching optimum

ENCODE implication: Adjust by error type and independence; test any numerical target

Desirable difficulty R31, R32

Proposed mechanism: Effort may support later learning despite lower current performance

Demonstrated outcome: Reliable for specific manipulations, not difficulty itself

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Difficulty is useful only if it produces better later learning

ENCODE implication: Judge effort by retention/transfer, not struggle or smoothness

Errorful learning R34

Proposed mechanism: Correct errors and update mistaken expectations

Demonstrated outcome: Errors with correction can support learning

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Uncorrected errors and persistent guessing can harm

ENCODE implication: Permit bounded attempts; preserve accurate examples and explicit repair

Errorless learning R15, R34

Proposed mechanism: Avoid competing wrong responses during acquisition

Demonstrated outcome: Accurate modeling can help novices; no universal error ban

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Clinical findings do not automatically generalize to this learner

ENCODE implication: Start complex procedures with correct models; still assess error diagnosis

Productive failure R35

Proposed mechanism: Generate attempts before consolidation/instruction

Demonstrated outcome: Problem-solving-first benefit under appropriate designs

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Age, content, prior knowledge, and instructional fidelity matter

ENCODE implication: Reserve for suitable concept comparisons; never equate abandonment with discovery

Transfer-appropriate practice R36, R37

Proposed mechanism: Practice operations that later tasks require

Demonstrated outcome: Retrieval transfer depends on relationships between practice and test

Grade for stated claim: MODERATE

Boundary conditions: Recall gains do not guarantee novel problem solving

ENCODE implication: Train and assess explanation, selection, construction, and application separately

Contextual variability R10, R21, R36

Proposed mechanism: Reduce reliance on superficial cues; compare invariants

Demonstrated outcome: Indirect, task-dependent support for varied cases

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Variation too early can hide the invariant; no arbitrary novelty rule

ENCODE implication: Change context/representation after structure is clear

Concept maps R38

Proposed mechanism: Externalize and construct relations

Demonstrated outcome: Meta-analysis supports average benefits over studied controls

Grade for stated claim: MODERATE

Boundary conditions: Constructing versus viewing contrasts may reflect other differences

ENCODE implication: Use small accurate relation maps; assess reconstruction and use, not map beauty

Overlearning R39, R40

Proposed mechanism: Additional practice after initial success

Demonstrated outcome: Retention benefits in some tasks; marginal value varies

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Massed extras can be less efficient than spacing

ENCODE implication: Stabilize safety/fluency-critical steps; move excess same-day practice to later sessions

Automaticity R41

Proposed mechanism: Repeated accurate component execution becomes faster

Demonstrated outcome: Supported theories and task evidence for practiced performance

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: Speed is task-specific and can conceal inflexibility

ENCODE implication: Train fluency after accuracy; retain exception and transfer checks

Confidence calibration R31, R32, R33

Proposed mechanism: Compare predicted and actual independent success

Demonstrated outcome: Familiarity and immediate fluency can mislead judgment

Grade for stated claim: STRONG for mismatch risk

Boundary conditions: Confidence is not competence; ratings alone do not fix it

ENCODE implication: Elicit confidence before correction and compare it with outcomes

Forgetting models R05, R57, R58

Proposed mechanism: Predict success as a function of practice history and delay

Demonstrated outcome: Useful task-specific empirical models

Grade for stated claim: MODERATE for prediction; HYPOTHESIS for ENCODE fit

Boundary conditions: Vocabulary fits do not determine procedure/explanation decay

ENCODE implication: Store observed delays first; fit and validate cautiously

Reconsolidation R42

Proposed mechanism: Proposed post-reactivation memory restabilization

Demonstrated outcome: Human evidence has important alternative explanations

Grade for stated claim: CONTEXT-DEPENDENT

Boundary conditions: No behavioral tutor observation establishes the mechanism

ENCODE implication: Use the term correction/representation revision operationally

Evidence tensions, not a universal recipe

Interleaving's overall synthesis estimate was $g = 0.42$, but benefits differed markedly by materials; word tasks favored blocking. Mixing topics simply to make study harder is not justified. R10

Self-explanation has broad support. However, the 2023 mathematics worked-example meta-analysis found smaller effects in studies with added self-explanation prompts than without them. This moderator does not prove the prompts caused harm. Earlier experiments also found benefits from well-targeted principle prompts with fading. The defensible policy is selective, checked explanation, followed by a component test. R20, R15, R18

Productive failure is a designed problem-solving-and-instruction sequence. Its synthesis found an average advantage for problem solving followed by instruction, with implementation and population boundaries. It does not authorize leaving a novice alone with an incomprehensible task. R35

4. Cognitive architecture and revision of the proposed pipeline

The initial pipeline conflates mechanisms, instructional actions, and endpoints. It should become a loop with multiple paths rather than a mandatory serial course.

Proposed component	Verdict	Scientific/operational replacement
Raw information	KEEP, add validation	Establish factual correctness, curriculum relevance, prerequisites, and assessable targets
Attention/orientation	KEEP, qualify	State the goal and focus relevant features; optionally prequestion. Do not claim measured neural attention
Initial encoding	KEEP as description	Provide an interpretable model/example or feasible activity; evaluate its result rather than announcing encoding
Mental representation/schema	KEEP as latent target	Infer relational/procedural structure from multiple tasks; revise it throughout learning
Guided retrieval	MODIFY	Distinguish independent attempts from hinted practice. Retrieval can recur early and later
Feedback/reconsolidation	RENAME	Corrective feedback and representation revision. Reconsolidation is not a required verified stage
Scaffold fading	KEEP conditionally	Remove support based on fresh performance; restore only the necessary support after failure

Proposed component	Verdict	Scientific/operational replacement
Fluency	MOVE to optional dimension	Important for bottleneck procedures, not a universal stage before conceptual transfer
Spaced reactivation	DISTRIBUTE throughout	Schedule knowledge and skill practice across sessions, conditional on goals and observed retention
Transfer	BEGIN earlier and reassess later	Include changed cases during acquisition; independently assess novel applications after delay

Functional architecture, HYPOTHESIS: validated source and assessment target feed a prerequisite diagnostic. The diagnostic selects an example, explanation, feasible prediction, completion task, or independent problem. Each response generates evidence: initial answer, reasoning, error signature, confidence, assistance, latency, and task features. Informative correction changes the next action. Independent fresh tasks check whether support can fade. A delayed review queue and separate transfer probes test durability and generalization. The system can return to instruction at any point.

Working memory and long-term knowledge interact; a set of elements difficult for a novice may function as a familiar unit for an expert. This motivates externalizing state and reducing unnecessary simultaneous demands. It does not justify measuring working-memory capacity from chat, inferring a brain disorder, or enforcing four bullets as a scientific rule. [R16](#), [R19](#)

Separate three system responsibilities: **content correctness**, **instructional policy**, and **assessment/state estimation**. A persuasive explanation cannot override an incorrect answer key. A state estimate cannot certify learning without an appropriate task. A policy's prediction of success does not establish that it causes learning.

The tutor should choose among a small set of interpretable actions. For each choice it should retain a short rationale, such as prerequisite failure, uncertain recall, confusable categories, excessive assistance, or an overdue delayed probe. This makes the future implementation auditable without pretending that the model can observe cognition directly.

5. ENCODE teaching loop v0

This is an exact candidate policy, not a proven optimal sequence. Its component rationale comes from the evidence map. Every count, time cap, and branch threshold here is **HYPOTHESIS** unless otherwise stated.

Sequence and branching logic

1. **Define one assessable target.** State what the learner should be able to do, under which tools/cues, and at what future delay. Select a narrow target that can be checked within the available session. Keep related context visible; one target does not mean severing its dependencies.
2. **Check the smallest relevant prerequisite.** Ask one short prediction, explanation, or execution question. If the learner already succeeds independently on the target and a changed case, skip introductory exposition and schedule delayed verification. If prerequisites fail, repair the specific gap first.
3. **Optionally pretest the target.** Allow one bounded guess if it is understandable and low cost. Record it as a diagnostic, not a failure penalty. Skip pretesting when even the notation or task meaning is unavailable. Provide the correct model soon afterward; do not repeatedly request unsupported answers.
4. **Teach one coherent unit.** Give a plain model, the exact term/notation, a correct worked example, and the crucial condition or contrast. For a complex procedure, expose the state/diagram and label functional subgoals. If the learner has relevant expertise, begin with a partial example or independent problem instead.
5. **Request a feasible independent response.** Remove the answer/example from the immediate response context when possible. Ask recall, a prediction, a next step, a trace, or a fresh mini-problem that matches the target. Do not ask “Do you understand?” as the learning check. Record confidence before correction on diagnostic/assessment probes, not necessarily every conversational turn.
6. **Classify the result provisionally.** Correct and adequately reasoned: move to a changed case. Correct but lucky, unexplained, or highly uncertain: probe the relevant relation or confusable alternative. Incorrect: use the repair branch below. Slow but correct: preserve the success and determine whether speed is actually a target.
7. **Vary and fade.** Change a surface feature, representation, condition, or task demand while preserving the taught structure. After two fresh independent successes with an adequate reason, remove one level of support. This provisional trigger must be tested; matching two memorized items does not qualify.
8. **Check independent application.** Give an unannounced fresh problem without the worked example, solution, or leading hint. Include method selection when relevant. If it fails, return to the specific weak dimension, not the beginning of the entire lesson.
9. **Close with a concise reconstruction.** Ask the learner to reconstruct the rule/procedure and its limit. Give a corrected compact reference afterward. Do not count a copied summary as independent recall. Record the next review and unresolved uncertainty.
10. **Reassess after delay.** Initially test at approximately 24 hours and seven days because these are practical measurement points, not optimal universal intervals. Use fresh parallel tasks. Schedule further practice according to goal horizon, observed failures, importance, and available time. A seven-day success does not establish month-long retention.

Incorrect-answer repair branch

First preserve the first attempt. Feedback must not overwrite evidence about what the learner could do unaided. Then identify the smallest plausible failure category. One response rarely proves the category; ambiguous cases require a second diagnostic.

Cannot recall; a neutral cue unlocks an accurate explanation/application

Provisional interpretation: Accessibility or cue failure

Next action: Supply the minimum cue, then use later free recall and a new context

What would disconfirm it: Continued errors despite a cue; inability to apply the idea

Recalls words but predicts the wrong outcome or condition

Provisional interpretation: Conceptual/relational misunderstanding

Next action: Show a contrasting case and explain the disputed relation; retry a changed case

What would disconfirm it: Correct independent reasoning on several fresh cases

Selects the right method but makes an execution error

Provisional interpretation: Procedure/component weakness

Next action: Model or complete the failing step; then execute independently

What would disconfirm it: Wrong method selection or a flawed explanation of the operation

Code logic is sound in a trace but syntax is invalid

Provisional interpretation: Syntax or representation failure

Next action: Supply the syntax reference/template; later test syntax without it if required

What would disconfirm it: Incorrect state changes even with valid syntax

Correct code cannot be run due to environment actions

Provisional interpretation: Tooling failure

Next action: Isolate and teach the environment action with a minimal known-good program

What would disconfirm it: Program itself is logically wrong or tool sequence is already correct

Confidently wrong on repeated changed cases

Provisional interpretation: Likely misconception, still provisional

Next action: Elicit prediction, show a diagnostic counterexample, explain correction, revisit after delay

What would disconfirm it: Error disappears on an independent diagnostic or scoring/key was wrong

Correct only after leading hints

Provisional interpretation: Scaffold-dependent performance

Next action: Practice at the needed support level, then fresh unaided probes

What would disconfirm it: Fresh independent success at the claimed dimension

Hint ladder: H0 = no hint; H1 = neutral retrieval cue; H2 = point to the relevant relation/subgoal; H3 = supply a partial step or completion structure; H4 = show and explain the worked solution. Offer only the next useful level. If the learner lacks the underlying model, go directly to H4 rather than perform a long hint ritual. If they request help or cannot make a feasible attempt, provide it. Record the greatest assistance level separately from correctness.

After correction, have the learner use the repaired idea on a fresh small item. An immediate repeat of the displayed answer is evidence of participation, not durable learning. Fade hints as independence improves; restore a targeted hint after failure. Treat persistent failure as a reason to inspect prerequisites, task validity, instructions, or overload, not to intensify guessing.

Adaptive decisions

- **Explain versus retrieve:** explain when notation, prerequisites, or the causal/execution model are missing; retrieve when a feasible response can reveal or strengthen knowledge. Explanation and practice should alternate. Neither continuous exposition nor continual testing is the default.
- **Repeat versus vary:** repeat the structural relation or difficult component; vary the case after it is interpretable. Repeated exact wording may improve wording recall without improving discrimination.
- **Block versus interleave:** give a novice enough same-family work to learn the operation. Mix confusable categories once the learner can attempt each, and measure whether selection improves. No fixed number of blocked items is established.
- **Difficulty:** maintain feasible effort and informative errors. The 85% rule is not a validated universal tutor target. Optional broad success bands should be tuned separately for recall, new procedures, and transfer, then validated locally. [R30](#)
- **Old material:** maintain a small due queue emphasizing important prerequisites, uncertain retention, and previous misconceptions. Expand intervals after independent delayed success; shorten or repair after failure. A proposed session budget is up to 20% review time, with urgent prerequisite repair exempt. This budget is a hypothesis, not a memory law.
- **Illusion of competence:** test without notes, without the familiar example, after delay, and on a changed problem. High confidence plus poor independent performance changes the instruction plan; high confidence alone does not.
- **Preference:** honor pace, concise wording, accessibility, and one main question at a time. Explain why a less comfortable task is useful when it tests a missing dimension. Offer a bounded choice about format; do not let preference remove the competence check or compel participation.

6. SLATE Text specification v0

SLATE should be a constrained instructional protocol, not a new vocabulary that replaces disciplinary language. Its purpose is to preserve necessary structure while reducing irrelevant processing and making retrieval demands explicit. No research located in this review establishes an optimal universal sentence length, exact chunk size, or synonym count for AI tutoring. The following language constraints are therefore design hypotheses except where narrower evidence is cited.

Structure of a teaching unit

Use: **target -> model -> exact term/notation -> mapped example -> condition/contrast -> one response request -> correction -> fresh use**. Omit already-known parts after diagnosis. A unit can span several turns; the sequence is not a demand to place all eight elements into one message.

Language/layout element	Specification	Evidence status and limit
Sentence size	Prefer one main relation or operation per sentence; retain a causal clause when it prevents ambiguity	HYPOTHESIS. No magic 12-, 20-, or 25-word limit

Language/layout element	Specification	Evidence status and limit
Chunk size	Begin with roughly 2-4 explanatory sentences per turn plus one relevant example/visual when needed; expand for coherent dependencies	HYPOTHESIS. Test by learning per minute, not brevity alone
Vocabulary	Use a stable label during initial acquisition. Define it once and map to standard variants later	HYPOTHESIS for synonym reduction; terminology consistency must not prevent transfer
Plain and exact language	Give a plain interpretation alongside the exact disciplinary term in the same initial unit	HYPOTHESIS for timing; exactness is required by the target, not postponed until after vague familiarity
Causal chain	Name the changed variable, each intermediate link, direction, and assumption; distinguish mechanism from correlation	Design requirement. Supported indirectly by coherent relational instruction, not a special SLATE causal law
Examples	Explicitly map example features to the rule. Include a boundary or non-example where confusion is likely	MODERATE/CONTEXT-DEPENDENT from case comparison and concreteness evidence
Compression	Compress after the learner can reconstruct the relation; keep a way to expand it	HYPOTHESIS. Shorter is not necessarily more efficient
Repetition	Repeat invariants through spaced retrieval, not identical rereading alone; change cases/representations deliberately	Evidence-backed components; their exact combination is HYPOTHESIS
Question granularity	One main response request per turn, with enough context to solve it; progressively include multi-step tasks	HYPOTHESIS personalized to the supplied preference

Language/layout element	Specification	Evidence status and limit
Scaffolding	Replace full causal frames with partial cues, then independent reconstruction as performance improves	MODERATE for fading; exact verbal templates are HYPOTHESIS
Formatting	Use short paragraphs, selective emphasis, aligned steps, and integrated labels; keep a visual near its explanation	MODERATE for relevant signaling/multimedia principles; no claim that formatting directly stores memory
Equations/code	Preserve exact symbols and indentation. Show how symbols map to quantities/state before manipulating them	Domain accuracy requirement; no audio-only or paraphrase-only substitute for precise notation

Concrete-first is optional. A learner who already understands the abstraction may benefit from an abstract rule followed by a case. A novice may need a meaningful example first, but the example must not become the entire concept. A colorful story that obscures the invariant is a poor example. Irrelevant interesting details can impair learning. [R21](#), [R22](#), [R44](#)

For IB, introduce the official term early and retain its distinctions. A plain paraphrase supports understanding; it should be followed by an accurate definition and tasks requiring the term in use. Verify the current syllabus, guide, and teacher markscheme before presenting wording as official. SLATE must not invent canonical definitions or collapse command terms into one generic answer pattern.

Illustrative causal unit, not an official IB quotation: “Assume income rises and the good is normal. At each price, buyers now want more of it. Demand shifts right. If supply is unchanged, the model predicts a higher equilibrium price and quantity.” A subsequent contrast asks what changes if the good is inferior. This preserves assumptions, direction, and the distinction between a shift and movement along a curve. It is not yet an assessment of what Advay has learned.

Progressive compression

Novice: complete relation and concrete mapping. Developing: partial chain with a missing reason or condition. Independent: terse cue requiring reconstruction. Experienced: compare cases, assumptions, and exceptions. Do not permanently train the learner to reproduce SLATE wording. Ask the same conceptual demand using standard textbook, examination, and natural-language variants after initial understanding.

Prior-knowledge interactions in text-coherence experiments also caution against a single optimal compression rule. Inferring omitted connections can help knowledgeable readers while making a novice's task harder. This is a boundary on simplification, not permission to deliberately make all expert instruction incoherent. [R59](#)

7. SLATE Voice specification v0

Speech is transient; dense instructions can disappear before the learner has coordinated their parts. Voice should therefore use learner-controlled pacing and a persistent visual surface when the content depends on exact external structure. This is a modality design issue, not evidence that Advay is an auditory or visual learner. [R26](#), [R23](#)

Candidate Voice rules:

1. Say the target and one main relation, then pause for processing or a response. An initial 20-40 second explanation segment is a testable interface heuristic, not a working-memory threshold.
2. Use explicit entity names when “it,” “this,” or “the second one” could become ambiguous. Restate the necessary condition before the question.
3. Avoid long serial lists, nested clauses, and reading full symbolic expressions aloud without visible notation. Show the expression and point to the relevant operation.
4. Keep code, equations, axes, tables, and state changes persistent and synchronized with speech. Use pauses to let the learner inspect them.
5. Ask one clear response request; allow thinking time without immediately supplying the answer. The tutor should not mistake silence for failure or fabricate comprehension from a short acknowledgment.
6. Offer pause, replay, and concise text recap. Captions/transcript are useful accessibility and review options; displaying and reading every word simultaneously is not always an optimal multimedia design.
7. After a wrong response, state the disputed relation and correction briefly. Then use a fresh task. Avoid a long corrective monologue containing several new concepts.
8. Move from spoken explanation to learner reconstruction and written execution where the actual goal requires writing, diagramming, or coding.

Audio-only exclusion for novice precision tasks, HYPOTHESIS grounded in task demands: do not teach new multi-line code, unfamiliar multi-step equations, coordinate diagrams, or detailed tables only through sound. Use a visible, accessible alternative. Audio can later support mental recall of a known formula, concept, or procedure, but that does not establish ability to write or execute it. If the learner is driving or otherwise cannot inspect a visual, restrict the task to appropriate verbal recall instead of pretending to teach detailed symbolic work.

8. Domain protocols

8.1 IB Economics and Business

Goal dimensions: exact terminology, causal explanation, diagram/model use, case application, and evaluation. Official Economics HL materials identify theory/model application and assessment demands; Business command terms and assessment requirements must be checked against the relevant guide. These documents establish curriculum targets, not the effectiveness of SLATE. [R60](#), [R61](#)

Candidate micro-lesson:

1. Diagnose the prerequisite distinction: for example change in demand versus change in quantity demanded, or revenue versus profit.
2. Introduce the term with exact distinctions and a plain meaning. Have the learner reconstruct it later without merely copying wording.
3. Build a causal chain with explicit assumptions. Include one case mapped to each link.
4. Construct the diagram or model in stages. Ask the learner to predict and label the next change, then produce a fresh diagram independently. Verbal recognition of a diagram is insufficient.
5. Present a contrasting case that changes a condition or exposes a confusable concept.
6. Apply to a new case with relevant evidence. Distinguish supplied case facts from invented examples.
7. Evaluate through mechanism, conditions, affected groups, time horizon, limitations, and a justified judgment. Use only factors relevant to the actual question; a generic checklist is not evaluation.
8. Assess without tutor cues, then revisit with changed cases and official task wording after delay.

For Business, emphasize decisions under case constraints, tradeoffs, calculations where required, and evidence supporting a recommendation. For Economics, emphasize model assumptions, variable relationships, shifts versus movements, diagram consistency, and qualified predictions. Both require explanations tied to the case rather than memorized paragraphs. Scaffold a short causal answer before requiring a full essay, but eventually assess integrated writing at the authentic target length.

Failure checks: correct definition but wrong causal sign; plausible chain without assumptions; memorized diagram with mislabeled axes; evaluation containing generic “depends” statements; evidence-free case claims; confident reproduction of an essay template without answering the question.

8.2 Mathematics

Goal dimensions: meaning of quantities and operations, valid transformations, method selection, procedural accuracy, discrimination, and transfer. Worked examples and spacing have useful mathematics-specific evidence; retrieval effects against restudy are less conclusively established in the 2025 synthesis. A real mathematical problem also requires construction and execution, not just recall of a formula. [R15](#), [R07](#)

Candidate micro-lesson: diagnose a prerequisite operation; map quantities/representation; study one correct worked example with a meaningful subgoal per group of steps; explain only a critical transformation; complete a missing step; solve a fresh same-family problem; contrast a tempting wrong method; mix with a confusable method; solve a changed representation or unfamiliar application; retest after delay. Keep equations visible. Fade step labels and templates as independent execution improves.

Use a little blocked work when the operation is new. Introduce interleaving when method selection is the target and the learner has enough knowledge to attempt the candidates. Randomly mixing unrelated topics is not equivalent to discriminative interleaving. A randomized classroom mathematics trial supports the practical value of mixed strategy-selection practice. [R11](#)

Classify errors by representation/model, method selection, transformation rule, arithmetic, notation, and checking. Ask for the point at which a worked solution first becomes invalid. Teach checks such as units, substitution, boundary values, or estimation only where they apply. Do not automatically respond to a conceptual error with more arithmetic drills.

Fluency practice begins after accurate execution and targets actual bottlenecks. Retain variation and exceptions so speed does not become brittle pattern matching. Avoid measuring reasoning quality through speed alone. Allocate some important practice to later days instead of spending all available time on surplus same-day repetitions. [R39](#), [R40](#)

8.3 Programming: execution before unsupported construction

The evidence supports scaffolds and explicit execution models more firmly than it supports one universally optimal progression. Tracing studies document novice difficulties but do not establish that a tracing-first curriculum causes better coding. PRIMM has encouraging school evidence; Parsons problems and subgoal examples have useful but heterogeneous findings. The ordered sequence below is an engineering synthesis to test, not a meta-analytically proven ladder. [R46](#), [R47](#), [R49](#), [R50](#)

SEE

Learner action: Inspect a short correct program and its input/output contract

Evidence required to move forward, proposed: Identify purpose, relevant variables, and known versus unknown constructs

Support removed: Irrelevant tooling/features

PREDICT/TRACE

Learner action: Predict output and update a visible state table step by step

Evidence required to move forward, proposed: Trace a fresh variant and locate the next executed statement

Support removed: Tutor-generated state updates

EXPLAIN

Learner action: Explain the mechanism or a crucial line in execution terms

Evidence required to move forward, proposed: Explain why a changed input or branch changes the result

Support removed: Full explanatory wording

MODIFY

Learner action: Make one meaningful change and predict consequences before running

Evidence required to move forward, proposed: Correct change plus matching prediction on a fresh case

Support removed: Exact edit instruction

COMPLETE

Learner action: Fill a missing line/subgoal; optionally arrange Parsons blocks

Evidence required to move forward, proposed: Complete an unseen variant and justify placement/logic

Support removed: Provided lines and ordering cues

GENERATE

Learner action: Write a tiny program/function from a plain contract

Evidence required to move forward, proposed: Independent code satisfying fresh tests and a trace/explanation

Support removed: Templates and model solution

DEBUG/TRANSFER

Learner action: Predict, observe, localize, hypothesize, test, repair; solve a new contract

Evidence required to move forward, proposed: Independent diagnosis and a justified repair; changed-context construction

Support removed: Tutor localization and proposed fix

Debugging begins early with one deliberately bounded discrepancy, and continues throughout. It should not wait until the learner finishes the entire ladder. Once a learner can trace and complete a tiny structure, begin small independent generation within the same lesson or soon afterward. Do not postpone all free coding for weeks, and do not begin with a large blank-page application. Prior independent evidence can bypass stages.

Execution model: distinguish source text, current statement, stored values/state, control flow, output, and errors. Use a language-accurate notional machine. Assignment is an operation, not an algebraic assertion; loops change state across iterations; function definitions and calls have different roles. These are examples of targets to verify in the selected language, not permission to generalize across all languages. [R47](#), [R48](#)

Keep syntax, logic, and tooling distinguishable during diagnosis. Use a stable minimal environment at first. Allow a syntax reference during logic practice if syntax recall is not the current target, but later assess without it when independent syntax production is required. Tool independence also needs explicit practice; hiding the environment forever would not teach real programming.

Subgoal labels should explain functions such as initialize state, iterate, update, and return, rather than decorate every line. A programming study found improved quizzes and fewer failures/withdrawals with labeled examples, but not a significant mean exam improvement. This boundary matters. [R51](#)

Use visualization to require prediction and inspection of state, not passive animation. Algorithm-visualization evidence emphasizes how learners engage with the representation. Debugging instruction should explicitly teach localization, hypothesis generation, controlled testing, and checking the repair rather than asking the AI for a corrected program. [R52](#), [R53](#)

Universal components versus domain adaptations

Universal candidates: correct targets, prerequisite diagnosis, interpretable examples, feasible independent practice, corrective information, reduced dependence, delayed checks, and explicit assessment of application. Domain-dependent elements: what counts as an explanation; diagram versus algebra versus runtime state; terminology fidelity; method selection; execution tools; appropriate fluency targets; and what novelty means. No evidence licenses transferring a factual-recall schedule unchanged to essay evaluation or debugging.

9. Multidimensional learner-state schema

Evidence sufficient for a learning claim

Use this statement format: “**At delay D, under cue/tool conditions C, on task family T, the learner independently demonstrated dimensions X.**” Do not say “mastered forever.” The following criteria are proposed minimum evidence gates, not universal psychometric cutoffs. Important claims require more samples and task diversity.

Dimension	Suitable observation	Insufficient evidence
Recognition	Select the target among plausible distractors	Seeing the answer and saying it looks familiar
Cued recall	Produce it from a specified partial cue without the answer	Completion after the tutor supplies the substantive answer
Free recall	Reconstruct core content with no answer-specific cue	Repeating wording still visible in the context
Conceptual explanation	Correct relations, conditions, and a suitable example, scored against a rubric	Fluent prose with missing or incorrect mechanism
Discrimination	Choose between confusable concepts/methods and justify the deciding feature	Isolated definition recall
Procedural execution	Perform fresh tasks accurately with permitted tools and no procedural hints	Following an existing solution
Application	Use the concept to explain or solve a new concrete case	Matching a nearly identical story by keywords
Near transfer	Independently handle a changed surface form within the trained structure	Repeated trained items
Novel/far transfer	Independently identify and use relevant structure in a predeclared, substantially changed task	Claiming general problem-solving improvement from one near-transfer item

Dimension	Suitable observation	Insufficient evidence
Speed/fluency	Accurate execution latency on comparable tasks, including variability	Fast incorrect answers or tutor-assisted speed
Confidence	A probability judgment before feedback on a specified response	Enthusiasm or retrospective certainty
Calibration	Compare confidence with independently scored outcomes across enough items	One confident correct answer
Retention after delay	Fresh independent performance at recorded elapsed time	Immediate performance or a prediction of future retention

For an initial learning claim, require at least two fresh independently solved items plus an appropriate explanation or discrimination check. For durability, repeat a parallel check at the intended delay. For transfer, include explicitly novel tasks. These small gates can detect obvious non-learning, but they cannot reliably estimate a stable success probability. Recognition, explanation, and transfer must remain separate even when correlated.

Observable event record

Store concept/skill IDs; prerequisites; source/key version; task ID and family; task novelty; targeted dimensions; actual prompt; item difficulty estimate and uncertainty; timestamp and elapsed delay; practice/exposure history; first answer; score and rubric; scoring uncertainty; error type; confidence before correction; hint sequence; highest assistance; answer exposure; active and elapsed time; tool conditions; interruption flags; and subsequent independent repair result. Preserve failed and incomplete attempts.

Concept-level representation, illustrative only

```
{
  "concept_id": "unassessed_concept",
  "content_version": null,
  "prerequisites": [{"id": "candidate_prerequisite", "status": "unverified"}],
  "dimensions": {
    "recognition": {"estimate": null, "uncertainty": "unmeasured", "evidence_ids": []},
    "cued_recall": {"estimate": null, "uncertainty": "unmeasured", "evidence_ids": []},
    "free_recall": {"estimate": null, "uncertainty": "unmeasured", "evidence_ids": []},
    "explanation": {"rubric_history": [], "evidence_ids": []},
    "discrimination": {"estimate": null, "evidence_ids": []},
    "procedure": {"estimate": null, "evidence_ids": []},
    "application": {"estimate": null, "evidence_ids": []},
    "near_transfer": {"estimate": null, "evidence_ids": []},
    "novel_transfer": {"estimate": null, "evidence_ids": []},
    "fluency": {"correct_latency_history": []},
    "confidence": {"prediction_history": []},
    "calibration": {"brier_score": null, "sample_count": 0},
    "delayed_retention": {"observed_probes": [], "forecast": null}
  },
  "misconceptions": [],
  "assistance_history": [],
  "next_action": {"type": "diagnose", "rationale": "no independent evidence"}
}
```

Null means unknown, not zero competence. No learner-specific numerical state has been fabricated here.

Latent state and updates

Maintain hypotheses about accessible knowledge, relational/procedural structure, misconceptions, and dependence. These are inferred variables, not directly observed traits. Conceptually update a belief with the likelihood of the response conditional on task features, delay, assistance, and scoring reliability: posterior belief is proportional to that likelihood times the prior belief. A guessed multiple-choice answer carries less evidence than independently generating and applying a correct rule. Hinted success updates assisted capability, not unassisted capability by the same amount.

Do not propagate a correct definition into a transfer score. Do not treat many nearly identical attempts as independent samples. A simple per-dimension evidence ledger is the appropriate initial baseline. A beta-binomial success summary may describe comparable binary probes, but naive use will understate uncertainty when items share cues or are dependent.

Knowledge-tracing models provide useful ideas about uncertainty, guess/slip effects, history, and skills, but answer prediction is not identical to multidimensional competence or causal instructional optimization. Standard latent-mastered/not-mastered assumptions may be too coarse here. Deep knowledge tracing is not justified by one learner's sparse trials. Compare interpretable baselines and calibrated probabilistic models only after adequate longitudinal data exist. [R58](#)

Retention and prerequisites

Begin with actual delayed probe results. If data later support forecasting, estimate success conditional on dimension, task/cue, prior successful retrievals, assistance, and elapsed time. Compare simple exponential or power-law candidates with empirical baselines on held-out future probes; do not present a fitted half-life as the brain's true decay constant. Vocabulary memory models demonstrate useful task-specific modeling, not a universal forgetting curve for explanations and procedures. [R57](#)

Prerequisite edges should be authored or validated from the domain and revised using diagnostics. A failure can raise uncertainty about a prerequisite; it does not automatically erase all descendants. Successful downstream performance can motivate checking an edge, but cannot establish all prerequisite knowledge. The graph should support action selection, not manufacture proof of learning.

Validate forecasts with temporal and item-family holdouts, calibration, proper scoring rules, and error analysis. Separate evaluation of prediction quality from randomized evaluation of learning policy. Better prediction of the next answer does not prove a better next lesson.

10. Failure-mode analysis

The primary adversary is a session that feels unusually clear while producing little later independence. Immediate fluency and confidence are particularly poor substitutes for delayed learning. [R31](#), [R33](#)

Overcompression before understanding

Observable warning: Learner repeats a slogan but cannot reconstruct links/conditions

Required countermeasure: Expand the model; require a new case and explanation

Residual risk: Excessive detail can also waste time

Passive familiarity

Observable warning: High confidence after reading; poor recall without context

Required countermeasure: Independent free recall and delayed checks

Residual risk: Testing itself changes learning

Retrieval too early

Observable warning: Repeated arbitrary guessing with unavailable prerequisites

Required countermeasure: Provide the missing model or a correct example

Residual risk: A brief pretest can still help orientation

Excessive retrieval

Observable warning: Same facts repeated while explanation/application remain weak

Required countermeasure: Sample different dimensions; compare time costs

Residual risk: More varied probes can introduce too much novelty

Hints reveal the answer

Observable warning: Correctness rises with H3/H4 support; unaided work stays poor

Required countermeasure: Preserve first attempts; fresh no-hint checks

Residual risk: Tutor wording can leak cues indirectly

Scaffold dependence

Observable warning: Learner succeeds only in the tutor's template

Required countermeasure: Fade and vary support; independent writing/construction

Residual risk: Abrupt removal may create avoidable failure

Memorized wording without schema

Observable warning: Definitions correct, predictions and contrasts wrong

Required countermeasure: Test relational meaning and conditions

Residual risk: Rubrics may reward polished language too much

Understanding without recall

Observable warning: Good explanation with notes; no accessible rule later

Required countermeasure: Spaced reconstruction and appropriate cues

Residual risk: Forgetting schedule may be uncertain

Recall without transfer

Observable warning: Facts retrieved, unfamiliar cases fail

Required countermeasure: Train and assess method selection/application

Residual risk: Far transfer may require untaught domain knowledge

Context-bound learning

Observable warning: Success depends on familiar phrasing/diagram layout

Required countermeasure: Vary cues, representation, and setting deliberately

Residual risk: Variation can confound item difficulty

Memorable but distorted example

Observable warning: Learner generalizes incidental story features

Required countermeasure: Map invariants; give a counterexample

Residual risk: Attractive examples may conceal bad mapping

Incorrect AI content/key

Observable warning: Reasonable learner answer marked wrong or false relation taught

Required countermeasure: Validate sources, worked solutions, and rubrics; track corrections

Residual risk: Verification can itself be fallible

Answer exposure before assessment

Observable warning: Learner sees a solution in chat/history during the check

Required countermeasure: Use a clean assessment context and independent response

Residual risk: Hidden recall of exact trained items remains possible

Confidence inflation

Observable warning: Confidence increases without delayed accuracy

Required countermeasure: Pre-feedback prediction and calibration display

Residual risk: Ratings add measurement burden

Overpersonalization

Observable warning: Preference is treated as a permanent cognitive identity

Required countermeasure: Start with general evidence; test differences locally

Residual risk: Small noisy trials can overfit personalization

Tool dependence

Observable warning: AI writes/runs code; learner cannot trace or construct independently

Required countermeasure: Separate supported practice from tool-free target checks

Residual risk: Real work legitimately uses tools; specify the target

Engagement substituted for learning

Observable warning: Streaks, praise, or session length dominate success reports

Required countermeasure: Optimize delayed competence and total time

Residual risk: Low burden and motivation still affect participation

Gamification misdiagnosed as inherently good/bad

Observable warning: Rewards chosen without learning outcomes

Required countermeasure: Treat optional rewards as a separate tested component

Residual risk: Meta-analytic gains do not prove every game mechanic works

Difficulty as theater

Observable warning: Learner struggles with little later benefit

Required countermeasure: Evaluate productive effort against delayed outcomes

Residual risk: Some worthwhile difficulties reduce immediate performance

Neural mechanism theater

Observable warning: Claims about dopamine, rewiring, brain programming, or reconsolidation without measurement

Required countermeasure: Use behavioral mechanisms and observable claims

Residual risk: Scientifically legitimate terms can still be overextended

Scoring contamination

Observable warning: Tutor judges its own coached wording as understanding

Required countermeasure: Blind scoring with predeclared rubrics and checked keys

Residual risk: LLM graders can share systematic bias

False durability

Observable warning: A same-session pass becomes a permanent mastery label

Required countermeasure: State task, cue, assistance, and delay explicitly

Residual risk: Sparse probes leave substantial uncertainty

Gamification has an evidence base with average learning benefits in some comparisons; rejecting engagement as a proxy is not the same as claiming gamification never works. Likewise, a structured AI tutor can perform well on immediate learning in a controlled setting. Neither establishes ENCODE's delayed benefit or its advantage over competent normal ChatGPT teaching. [R45](#), [R55](#)

11. Personalized N-of-1 experimental program

Purpose and estimands

Determine whether an ENCODE protocol improves Advay's independent delayed learning per unit of total study time, and which components cause the improvement. These experiments cannot establish a population effect or validate a fixed learner identity.

Primary comparison: high-quality normal ChatGPT teaching (A) versus the explicit ENCODE/SLATE bundle (B), using the same model/version, subject resources, correctness checks, prior-knowledge information, and learning target. A should be a strong, responsive tutor capable of examples, explanation, practice, and correction. It must not be forced into passive lectures or denied proven techniques to make B look better. B applies the proposed diagnostic, representation, retrieval, fading, and delayed-review policy. Differences and tutor transcripts must be logged.

Estimate two complementary quantities: (1) independent delayed performance with a fixed teaching-time budget; (2) total active learning and review time to reach a predeclared performance criterion. Time to an immediate criterion alone is not sufficient. Select one primary performance outcome per experiment; retain the remaining dimensions as separate secondary outcomes.

Crossover without an impossible washout

Learning is not readily reversible. Teaching the same concept with A and then B does not create an unbiased crossover: B inherits A's learning. No washout period can be assumed to erase knowledge. Instead, **cross methods across matched fresh learning units**, counterbalance AB/BA order, and use new task banks. This retains within-person control while acknowledging unequal topics and carryover.

N-of-1 reporting principles such as prespecification, allocation, outcome timing, and transparent treatment history are useful analogies from the CENT statement. The clinical trial framework is not itself validation of an educational crossover. [R56](#)

Practical stages

Stage	Design	Purpose and decision
Baseline and materials	Independent short diagnostics by domain; verify item keys and prerequisite assumptions	Establish usable starting points; no strength ranking from impressions
Feasibility pilot	Two matched A/B pairs per domain: six pairs, twelve learning units	Test timing, scoring, floor/ceiling effects, compliance, and task equivalence; do not claim efficacy
Main bundle study	Four matched A/B pairs per domain: twelve pairs, twenty-four fresh units, balanced AB/BA	Estimate domain-specific delayed outcomes and time; do not pool incompatible raw scores
Confirmation	Approximately ten further matched pairs in the most promising domain, with new items	Check whether the pattern replicates; sample size is provisional, not a power guarantee
Component studies	Fresh matched units comparing one component while keeping the rest fixed	Identify useful mechanisms instead of attributing every gain to SLATE

An initial unit might use a 15-minute teaching cap plus about six minutes of assessment at 24 hours and seven days. For twenty-four units this is approximately 10.8 hours, excluding baseline/material preparation and any separate immediate-assessment time. Spread units and delayed checks over four to six weeks; revise burden after the pilot. Record active learner time and elapsed session time separately, including tutor reading, hint use, correction, and reviews. System-generation waiting time should not be silently counted as efficient learning or removed from user-experienced cost.

Small experiments across domains

Economics/Business bundle

Candidate contrast: A versus B

Fresh-unit target: Matched unfamiliar micro-concepts with definition, causal case, and conditional judgment

Primary outcome: Seven-day independently scored case application/evaluation

Important limitation: Match difficulty and prior knowledge; concept pairs must not teach one another

Economics/Business component

Candidate contrast: Concise causal chain plus mapped contrast versus equal-content competent prose/examples

Fresh-unit target: Conditions and confusable concepts

Primary outcome: Delayed discrimination plus explanation

Important limitation: “Concise” must not change factual coverage; test template dependence

Mathematics bundle

Candidate contrast: A versus B

Fresh-unit target: Matched procedures with conceptual explanation and method selection

Primary outcome: Seven-day independent procedural/application score

Important limitation: Procedural families differ; establish prerequisites first

Mathematics component

Candidate contrast: Targeted self-explanation prompt versus no additional prompt with the same worked example/fading

Fresh-unit target: One critical transformation

Primary outcome: Delayed execution/transfer per total minute

Important limitation: Tests a live evidence tension; do not require self-explanation in both arms

Programming bundle

Candidate contrast: A versus execution-model/trace/modify/complete/generate B

Fresh-unit target: Matched tiny programs using familiar tooling

Primary outcome: Delayed independent generation and debugging

Important limitation: Code length alone does not match complexity; syntax and logic scored separately

Programming component

Candidate contrast: Completion/Parsons bridge versus an equally supported small generation bridge

Fresh-unit target: A new control/data structure with shared prerequisites

Primary outcome: Independent construction after delay

Important limitation: Matching time and assistance is essential; no large blank-page strawman

Choose concrete units only after baseline. Do not select an apparently “weak” math topic or “strong” economics topic from conversation impressions. A domain expert should construct paired banks containing trained, immediate, delayed, near-transfer, and novel-transfer items. Pilot representative items for gross difficulty and scoring problems. Avoid reusing the same question with different numbers as the only transfer check.

Allocation and contamination control

Randomize which method receives which member of a pair, and counterbalance session order within domain. Generate and record allocation before instruction. Keep immediate/delayed assessors blind to condition where practical. Match model/version, resource access, key accuracy, session cap, and allowed tools. The tutor sees prior diagnostics but must not see held-out assessment answers.

Record sleep/time of day only as contextual covariates if practical, without attributing medical meaning. Record other study, interruptions, accidental answer exposure, method deviations, and changes to the AI model. Learning to use ENCODE can carry over to A; time/order effects should be reported, not assumed absent. New topic banks and counterbalancing reduce, but cannot eliminate, these problems.

Measurement schedule and rubrics

Measure immediate free recall **before** recognition or answer-bearing cues. Then measure explanation, discrimination, execution/application as appropriate, confidence, and assistance. Use parallel forms at 24 hours and approximately seven days; record actual elapsed hours. Include near and novel transfer at the delayed checkpoint. Exact timing windows, initially 24 hours plus/minus four hours and seven days plus/minus one day, are practical hypotheses.

Measure	Prespecified scoring
Time to initial criterion	Time until two fresh independent successes plus required explanation/condition; failed-to-reach criterion remains a censored failure at the cap
Free recall	Essential proposition/step coverage with penalties for substantive false relations; no credit merely for matching SLATE wording
Explanation	Four components scored 0-2 each: correct relation/mechanism, conditions, concrete mapping, and justified prediction; total 0-8
Discrimination	Correct choice plus the deciding feature; assess both familiar and changed contexts
Procedure	Required steps/output, critical error count, and independence; define critical errors before teaching
Near/novel transfer	Task-specific rubric with novelty dimensions declared in advance, not labeled “far” after a favorable result
Programming	Separate execution model, algorithm/logic, syntax, testing/debugging, and tooling; fresh tests alone do not establish explanation
Response speed	Correct-response latency on comparable items; report typing/tool overhead and speed-accuracy tradeoffs
Confidence/calibration	Probability of correctness before feedback; Brier score = mean of (predicted probability minus binary outcome) squared for binary-scored items
Assistance	First-attempt accuracy, maximum hint level, answer exposure, and count of independent successes
Retention	Performance at the actual delay under recorded intervening practice, not an inferred percentage of memory remaining

For partial-credit tasks, preserve component scores and use appropriately defined prediction errors; do not call a confidence-versus-continuous-rubric discrepancy a standard binary Brier score. Confidence intervals and calibration plots will be unstable with few observations.

A tutor-independent grader should use prewritten rubrics and checked keys. Blind transcript condition/style where possible. Review a sample with a teacher or competent human, especially cases, explanations, and novel programming solutions. If only AI grading is available, compare independent grading passes, adjudicate disagreements, and disclose remaining uncertainty. Do not let the teaching model grade its own explanation quality as the primary outcome.

Testing changes learning. The 24-hour test is a reactivation that can improve the seven-day result. Equalize assessment and feedback between arms and describe seven-day retention as retention under that

measurement regime. If estimating unreactivated seven-day retention matters, allocate some parallel items to a seven-day-only probe. Log all intervening study; do not claim a natural forgetting rate from repeatedly tested items.

Analysis and criteria for changing ENCODE

Keep paired differences for each domain and outcome. Plot all pairs, including failures and incomplete units. Report time distributions and the proportion reaching criterion, not just successful-session averages. Use randomization-based paired analysis where allocation supports it; uncertainty from small samples remains wide. Any bootstrap should resample pairs, not individual responses that share a lesson. More elaborate hierarchical analysis is optional after enough data, not a prerequisite for the pilot.

Predeclare missing-data handling. A reached time cap is failure to reach the criterion, not a missing success time. A missed follow-up is missing retention evidence, not automatically zero and not evidence of success. Report available paired cases plus sensitivity bounds for missing outcomes. Avoid selecting whichever of thirteen measures looks favorable.

Provisional utility thresholds, HYPOTHESIS: a useful protocol might save at least 20% of total active learning/review time while keeping seven-day independent application/transfer within a predeclared five-percentage-point non-inferiority margin; alternatively it might improve that delayed score by at least ten percentage points at similar time. These are proposed practical values, not empirical laws. They need a scoring scale with enough resolution and explicit uncertainty. A small sample that cannot exclude meaningful harm does not establish non-inferiority.

Change the default only when the direction is reasonably consistent across matched units, independence is preserved, delayed outcomes support the change, and new-item confirmation repeats the pattern. If uncertainty spans benefit and harm, keep the change optional and gather more evidence. A large gain in one domain supports a domain-specific choice, not a universal learner rule. Repeated failures, heavier hint dependence, or a delayed-transfer decrement justify revising B even if it is preferred or feels clear.

12. Open scientific questions

1. Does the bundle beat high-quality normal ChatGPT once factual accuracy, baseline knowledge, time, and follow-up testing are equalized?
2. Does concise presentation save time without sacrificing conditions, relational understanding, and novel transfer? Where is its compression limit?
3. Does one main question per turn help enough to offset extra conversational overhead? When should integrated multi-step work begin?
4. Which self-explanation prompts improve Advay's learning, and which merely increase time or unsupported narration?
5. When does a programming completion/Parsons bridge outperform early tiny-program generation, and when does it delay independence?
6. What amount and kind of initial blocked practice makes subsequent interleaving useful in each domain?
7. Which cue and hint policies strengthen later recall rather than teach dependence? Can indirect answer leakage be detected reliably?
8. How much transfer requires additional instruction rather than repeated retrieval of the original rule?
9. Can separate retention forecasts for concepts, procedures, and explanation be estimated from a feasible amount of data?

10. Does SLATE Voice improve access or time at equal delayed performance, and which persistent visual supports are necessary?
11. Can misconception diagnosis be made reliably from a few conversational responses without overinterpreting wording?
12. Which personalized adaptations replicate across new topics, and which are local interactions with task or prior knowledge?
13. What is the value of a concept graph beyond a straightforward prerequisite list and event ledger?
14. How should total review burden be allocated when many topics compete and time is limited?
15. Do benefits persist beyond seven days, across examination-style tasks, and after the learner stops using the tutor?

13. ENCODE v0 scientific constitution

Grades apply to the stated scientific claim; translating it into software remains an engineering decision. The thresholds and sequencing used to implement a rule are not upgraded to STRONG by a strong underlying phenomenon.

1. **STRONG:** Evaluate learning through independent performance, including delay where durability is claimed. Immediate fluency is insufficient. [R31](#), [R32](#)
2. **STRONG:** Include retrieval opportunities for important knowledge, with corrective information; do not infer superiority over every active comparator or every domain. [R02](#), [R03](#), [R04](#), [R07](#)
3. **STRONG:** Distribute practice across time when retention matters; choose intervals from goals and observations, not a universal calendar. [R05](#), [R06](#), [R07](#)
4. **STRONG:** Give novices accurate, interpretable worked examples for complex procedures; verify that the learner understands the relevant structure. [R15](#)
5. **STRONG:** Provide actionable corrective information and retain the initial unassisted response as evidence. Feedback type and timing still require adaptation. [R27](#), [R28](#)
6. **MODERATE:** Re-establish successful retrieval across spaced sessions for important learnable knowledge; test efficiency for non-factual tasks. [R08](#), [R09](#)
7. **MODERATE:** Fade examples and hints toward independent construction; require fresh evidence before claiming independence. [R17](#), [R18](#)
8. **CONTEXT-DEPENDENT:** Interleave to practice discrimination and strategy selection when prerequisites permit; do not mix solely to create difficulty. [R10](#), [R11](#)
9. **CONTEXT-DEPENDENT:** Use targeted self-explanation when it addresses a learning bottleneck; avoid compulsory explanations of every step. [R20](#), [R15](#), [R18](#)
10. **MODERATE:** Map examples and contrasting cases to their invariant structure and limits. [R21](#), [R22](#)
11. **CONTEXT-DEPENDENT:** Choose complementary representations, signaling, and pacing for the task; persistent symbols matter when speech is transient. [R23](#), [R24](#), [R25](#), [R26](#)
12. **MODERATE:** Teach a language-accurate execution model and scaffold novice code work; the exact SEE-to-GENERATE progression remains a hypothesis. [R46](#), [R47](#), [R49](#), [R50](#)
13. **MODERATE:** Teach debugging as diagnosis and testing, and assess it independently. [R53](#)
14. **STRONG:** Treat confidence and familiarity as evidence about metacognition, not as demonstrated competence. [R31](#), [R32](#), [R33](#)

15. **CONTEXT-DEPENDENT:** Assess transfer directly; vary appropriate cues and tasks without assuming a general far-transfer benefit. [R36](#), [R37](#)
16. **HYPOTHESIS:** Use concise SLATE units, stable initial labels, and one main question per turn for Advay, then retain them only if outcomes support them.
17. **HYPOTHESIS:** Maintain multidimensional state with explicit uncertainty, assistance, and task/delay conditions; validate forecasts separately from instructional effects. [R57](#), [R58](#)
18. **HYPOTHESIS:** Optimize total time subject to delayed independent outcome requirements; reject any adaptation that merely increases assisted fluency or confidence. [R54](#), [R55](#)

14. Bibliography and source-inspection register

DOI links identify the academic work even when full text requires access. F/A/O codes report this review's inspection, not source quality. Author manuscripts/repository abstracts were used where publisher access failed. Several reviews share primary studies; their samples and effects must not be counted as independent cumulative evidence.

R01. Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). *Improving Students' Learning With Effective Learning Techniques: Promising Directions From Cognitive and Educational Psychology*. Psychological Science in the Public Interest, 14(1), 4-58. DOI. Broad critical review; **A**, publisher/academic summary. Used for relative utility and elaboration boundaries.

R02. Yang, C., Luo, L., Vadillo, M. A., Yu, R., & Shanks, D. R. (2021). *Testing (quizzing) boosts classroom learning: A systematic and meta-analytic review*. Psychological Bulletin, 147(4), 399-435. DOI. Meta-analysis; **A**, PubMed abstract. Used for classroom testing and moderators.

R03. Agarwal, P. K., Nunes, L. D., & Blunt, J. R. (2021). *Retrieval Practice Consistently Benefits Student Learning: a Systematic Review of Applied Research in Schools and Classrooms*. Educational Psychology Review, 33, 1409-1453. DOI. Systematic review; **A**, publisher abstract. Applied evidence has limited representation outside commonly studied populations.

R04. Gonçalves, A. de O., Muniz, B. F. B., & Jaeger, A. (2025). *Retrieval Practice Versus Elaborative Encoding: A Systematic and Meta-analytic Review*. Educational Psychology Review, 37, article 100. DOI. Meta-analysis; **A**, publisher abstract. Important stronger-comparator and feedback boundary.

R05. Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). *Distributed practice in verbal recall tasks: A review and quantitative synthesis*. Psychological Bulletin, 132(3), 354-380. DOI. Meta-analysis; **A**, academic repository/abstract. [Repository](#). Verbal recall and retention-interval dependence.

R06. Mawson, R. D., & Kang, S. H. K. (2025). *The Distributed Practice Effect on Classroom Learning: A Meta-Analytic Review of Applied Research*. Behavioral Sciences, 15(6), 771. DOI. Meta-analysis; **A**, PubMed abstract. Classroom spacing rather than a precise individualized schedule.

R07. Murray, E., Horner, A. J., & Göbel, S. M. (2025). *A Meta-analytic Review of the Effectiveness of Spacing and Retrieval Practice for Mathematics Learning*. Educational Psychology Review, 37, article 75. DOI. Meta-analysis; **A**, university repository abstract. [Repository](#). Mathematics spacing supported; retrieval contrast inconclusive.

R08. Rawson, K. A., Dunlosky, J., & Sciarrelli, S. M. (2013). *The Power of Successive Relearning: Improving Performance on Course Exams and Long-Term Retention*. Educational Psychology Review, 25, 523-548. DOI. Classroom experiments; **A**, ERIC abstract. Important knowledge relearned across sessions.

R09. Rawson, K. A., & Dunlosky, J. (2022). *Successive Relearning: An Underexplored but Potent Technique for Obtaining and Maintaining Knowledge*. Current Directions in Psychological Science. DOI. Review; **A**, publisher indexed article excerpts. Does not establish a universal procedure schedule.

- R10.** Brunmair, M., & Richter, T. (2019). *Similarity matters: A meta-analysis of interleaved learning and its moderators*. *Psychological Bulletin*, 145(11), 1029-1052. DOI. Meta-analysis; **A**, PubMed abstract. Material- and similarity-dependent effects.
- R11.** Rohrer, D., Dedrick, R. F., Hartwig, M. K., & Cheung, C. N. (2020). *A randomized controlled trial of interleaved mathematics practice*. *Journal of Educational Psychology*, 112(1), 40-52. DOI. Classroom randomized trial; **A**, academic/publisher record. Strategy selection, not arbitrary topic switching.
- R12.** Bertsch, S., Pesta, B. J., Wiscott, R., & McDaniel, M. A. (2007). *The generation effect: A meta-analytic review*. *Memory & Cognition*, 35, 201-210. DOI. Meta-analysis; **A**, publisher abstract. Generation memory tasks differ from whole-program construction.
- R13.** Pan, S. C., & Carpenter, S. K. (2023). *Prequestioning and Pretesting Effects: a Review of Empirical Research, Theoretical Perspectives, and Implications for Educational Practice*. *Educational Psychology Review*, 35, article 97. DOI. Review; **F**, relevant open-access sections. Correction, attention, and generalization boundaries.
- R14.** St. Hilaire, K. J., Chan, J. C. K., & Ahn, D. (2024; online 2023). *Guessing as a learning intervention: A meta-analytic review of the prequestion effect*. *Psychonomic Bulletin & Review*, 31, 411-441. DOI. Meta-analysis; **A**, publisher abstract. Specific versus general prequestion effects.
- R15.** Barbieri, C. A., Miller-Cotto, D., Clerjuste, S. N., & Chawla, K. (2023). *A Meta-analysis of the Worked Examples Effect on Mathematics Performance*. *Educational Psychology Review*, 35, article 11. DOI. Meta-analysis; **F**, author PDF abstract and targeted results. Prompt moderator must not be interpreted as isolated causal proof.
- R16.** Sweller, J., van Merriënboer, J. J. G., & Paas, F. (2019). *Cognitive Architecture and Instructional Design: 20 Years Later*. *Educational Psychology Review*, 31, 261-292. DOI. Theoretical/instructional review; **A**, publisher abstract/indexed sections. Load theory is not a measured chat-based brain capacity.
- R17.** Renkl, A., & Atkinson, R. K. (2003). *Structuring the Transition From Example Study to Problem Solving in Cognitive Skill Acquisition: A Cognitive Load Perspective*. *Educational Psychologist*, 38(1), 15-22. DOI. Review; **A**, academic record. Rationale for fading and completion problems.
- R18.** Atkinson, R. K., Renkl, A., & Merrill, M. M. (2003). *Transitioning From Studying Examples to Solving Problems: Effects of Self-Explanation Prompts and Fading Worked-Out Steps*. *Journal of Educational Psychology*, 95(4), 774-783. DOI. Controlled experiments; **A**, institutional abstract. Specific prompted-fading results, not a universal sequence.
- R19.** Cowan, N. (2010). *The Magical Mystery Four: How is Working Memory Capacity Limited, and Why?* *Current Directions in Psychological Science*, 19(1), 51-57. DOI. Review; **A**, academic record. Does not establish an optimal number of instructional bullets.
- R20.** Bisra, K., Liu, Q., Nesbit, J. C., Salimi, F., & Winne, P. H. (2018). *Inducing Self-Explanation: A Meta-Analysis*. *Educational Psychology Review*, 30, 703-725. DOI. Meta-analysis; **A**, ERIC abstract. Broad synthesis with task and prompt heterogeneity.
- R21.** Fyfe, E. R., McNeil, N. M., Son, J. Y., & Goldstone, R. L. (2014). *Concreteness Fading in Mathematics and Science Instruction: a Systematic Review*. *Educational Psychology Review*, 26, 9-25. DOI. Systematic review; **A**, ERIC abstract. Mapping concrete to abstract is conditional.
- R22.** Alfieri, L., Nokes-Malach, T. J., & Schunn, C. D. (2013). *Learning Through Case Comparisons: A Meta-Analytic Review*. *Educational Psychologist*, 48(2), 87-113. DOI. Meta-analysis; **F**, author PDF targeted sections. Supports comparison with explicit structural mapping.
- R23.** Noetel, M., et al. (2022). *Multimedia Design for Learning: An Overview of Reviews With Meta-Meta-Analysis*. *Review of Educational Research*, 92(3), 413-454. DOI. Overview of reviews; **A**, academic/publisher abstract. Overlap and multiple comparators limit effect aggregation.

- R24.** Rey, G. D., Beege, M., Nebel, S., Wirzberger, M., Schmitt, T. H., & Schneider, S. (2019). *A Meta-analysis of the Segmenting Effect*. *Educational Psychology Review*, 31, 389-419. DOI. Meta-analysis; A, publisher abstract. Learning benefit and added study time both matter.
- R25.** Schneider, S., Beege, M., Nebel, S., & Rey, G. D. (2018). *A meta-analysis of how signaling affects learning with media*. *Educational Research Review*, 23, 1-24. DOI. Meta-analysis; A, publisher abstract. Relevant signaling rather than indiscriminate highlighting.
- R26.** Leahy, W., & Sweller, J. (2011). *Cognitive load theory, modality of presentation and the transient information effect*. *Applied Cognitive Psychology*, 25(6), 943-951. DOI. Experiments; A, academic abstract. Duration and persistent information matter for voice design.
- R27.** Wisniewski, B., Zierer, K., & Hattie, J. (2020). *The Power of Feedback Revisited: A Meta-Analysis of Educational Feedback Research*. *Frontiers in Psychology*, 10, 3087. DOI. Meta-analysis; A, publisher abstract/results summary. Substantial heterogeneity by feedback information.
- R28.** Shute, V. J. (2008). *Focus on Formative Feedback*. *Review of Educational Research*, 78(1), 153-189. DOI. Review; A, publisher/academic record. Timing and hint policy are conditional.
- R29.** Kulik, C.-L. C., Kulik, J. A., & Bangert-Drowns, R. L. (1990). *Effectiveness of Mastery Learning Programs: A Meta-Analysis*. *Review of Educational Research*, 60(2), 265-299. DOI. Meta-analysis; A, publisher abstract. Program effects do not determine a universal mastery percentage.
- R30.** Wilson, R. C., Shenhav, A., Straccia, M., & Cohen, J. D. (2019). *The Eighty Five Percent Rule for optimal learning*. *Nature Communications*, 10, 4646. DOI. Computational/theoretical study; F, scope and model sections. Not a validation of 85% accuracy in general human tutoring.
- R31.** Soderstrom, N. C., & Bjork, R. A. (2015). *Learning versus performance: an integrative review*. *Perspectives on Psychological Science*, 10(2), 176-199. DOI. Review; A, academic abstract. Current performance and durable learning can diverge.
- R32.** Bjork, R. A., Dunlosky, J., & Kornell, N. (2013). *Self-Regulated Learning: Beliefs, Techniques, and Illusions*. *Annual Review of Psychology*, 64, 417-444. DOI. Review; A, publisher record/abstract. Metacognitive illusions and theoretical retrieval/storage distinctions.
- R33.** Benjamin, A. S., Bjork, R. A., & Schwartz, B. L. (1998). *The mismeasure of memory: when retrieval fluency is misleading as a metamnemonic index*. *Journal of Experimental Psychology: General*, 127(1), 55-68. DOI. Experiments; A, academic/PubMed record. Retrieval fluency can mislead memory judgments.
- R34.** Metcalfe, J. (2017). *Learning from Errors*. *Annual Review of Psychology*, 68, 465-489. DOI. Review; A, publisher abstract. Errors with correction, not unbounded guessing.
- R35.** Sinha, T., & Kapur, M. (2021). *When Problem Solving Followed by Instruction Works: Evidence for Productive Failure*. *Review of Educational Research*, 91(5), 761-798. DOI. Meta-analysis; A, publisher abstract. Implementation and population conditions matter.
- R36.** Pan, S. C., & Rickard, T. C. (2018). *Transfer of test-enhanced learning: Meta-analytic review and synthesis*. *Psychological Bulletin*, 144(7), 710-756. DOI. Meta-analysis; F, author PDF targeted abstract/transfer discussion. No blanket far-transfer guarantee.
- R37.** Morris, C. D., Bransford, J. D., & Franks, J. J. (1977). *Levels of processing versus transfer appropriate processing*. *Journal of Verbal Learning and Verbal Behavior*, 16(5), 519-533. DOI. Experiments; A, publisher abstract. Practice/test processing correspondence.
- R38.** Schroeder, N. L., Nesbit, J. C., Anguiano, C. J., & Adesope, O. O. (2018; online 2017). *Studying and Constructing Concept Maps: a Meta-Analysis*. *Educational Psychology Review*, 30, 431-455. DOI. Meta-analysis; A, ERIC abstract. Construction-versus-viewing subgroup differences need caution.

- R39.** Driskell, J. E., Willis, R. P., & Copper, C. (1992). *Effect of overlearning on retention*. Journal of Applied Psychology, 77(5), 615-622. DOI. Meta-analysis; A, academic abstract. Task and retention interval moderate utility.
- R40.** Rohrer, D., & Taylor, K. (2006). *The effects of overlearning and distributed practise on the retention of mathematics knowledge*. Applied Cognitive Psychology, 20(9), 1209-1224. DOI. Experiments; A, publisher abstract. Extra massed work and spacing are distinct allocation choices.
- R41.** Logan, G. D. (1988). *Toward an instance theory of automatization*. Psychological Review, 95(4), 492-527. DOI. Theory with empirical basis; A, academic abstract. Automaticity is not equivalent to flexible conceptual competence.
- R42.** Elsey, J. W. B., van Ast, V. A., & Kindt, M. (2018). *Human memory reconsolidation: A guiding framework and critical review of the evidence*. Psychological Bulletin, 144(8), 797-848. DOI. Critical review; A, academic abstract. Mechanism inference has substantial limits.
- R43.** Pashler, H., McDaniel, M., Rohrer, D., & Bjork, R. (2008; online 2009). *Learning Styles: Concepts and Evidence*. Psychological Science in the Public Interest, 9(3), 105-119. DOI. Critical review; A, publisher abstract/indexed conclusions. Preference does not validate style-matched instruction.
- R44.** Sundararajan, N., & Adesope, O. (2020). *Keep it Coherent: A Meta-Analysis of the Seductive Details Effect*. Educational Psychology Review, 32, 707-734. DOI. Meta-analysis; A, publisher abstract. Interesting irrelevant material can impair learning.
- R45.** Sailer, M., & Homner, L. (2020; online 2019). *The Gamification of Learning: a Meta-analysis*. Educational Psychology Review, 32, 77-112. DOI. Meta-analysis; A, publisher abstract. Learning outcomes differ from engagement metrics.
- R46.** Lister, R., et al. (2004). *A multi-national study of reading and tracing skills in novice programmers*. ITiCSE working group reports, 119-150. DOI. Multi-institution assessment study; A, ACM/academic record. Diagnoses tracing gaps; does not randomize curriculum order.
- R47.** Sorva, J. (2013). *Notional Machines and Introductory Programming Education*. ACM Transactions on Computing Education, 13(2), article 8. DOI. Review/conceptual synthesis; A, institutional record. Explicit execution models for introductory programming.
- R48.** Qian, Y., & Lehman, J. (2017). *Students' Misconceptions and Other Difficulties in Introductory Programming: A Literature Review*. ACM Transactions on Computing Education, 18(1). DOI. Literature review; F, author-uploaded targeted sections. Misconception categories are not diagnoses of Advay.
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- R50.** Sentance, S., Waite, J., & Kallia, M. (2019). *Teaching computer programming with PRIMM: a sociocultural perspective*. Computer Science Education, 29(2-3), 136-176. DOI. School intervention with controls/mixed methods; A, university abstract. Classroom social components may not transfer unchanged to one-to-one AI teaching.
- R51.** Margulieux, L. E., Morrison, B. B., & Decker, A. (2020). *Reducing withdrawal and failure rates in introductory programming with subgoal labeled worked examples*. International Journal of STEM Education, 7, article 19. DOI. Course intervention; F, open-access abstract/results. Quiz and withdrawal/failure results differ from mean exam results.
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- R53.** Yang, S., Baird, M., O'Rourke, E., Brennan, K., & Schneider, B. (2024). *Decoding Debugging Instruction: A Systematic Literature Review of Debugging Interventions*. ACM Transactions on Computing Education, 24(4), article 45. DOI. Systematic review; F, publisher review/intervention sections. Diverse studies and outcome measures limit universal prescriptions.

- R54.** Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakçı, Ö., & Mariman, R. (2025). *Generative AI without guardrails can harm learning: Evidence from high school mathematics*. PNAS, 122(26), e2422633122. DOI. Randomized field study; **F**, relevant published abstract/design/results. Guarded tutor avoided harm but did not establish a significant unassisted advantage. Published correction concerns affiliation, not the reported outcome.
- R55.** Kestin, G., Miller, K., Klaes, A., Milbourne, T., & Ponti, G. (2025). *AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting*. Scientific Reports, 15, 17458. DOI. Randomized physics instruction study; **A**, publisher abstract/design summary. Immediate outcomes and comparator setting do not establish long-term ENCODE efficacy.
- R56.** Vohra, S., et al., CENT Group (2015). *CONSORT extension for reporting N-of-1 trials (CENT) 2015 Statement*. BMJ, 350, h1738. DOI. Reporting guidance; **A**, published abstract/checklist excerpts. Production correction. Applied here as a methodological analogy, not an educational efficacy study.
- R57.** Pavlik, P. I., Jr., & Anderson, J. R. (2005). *Practice and forgetting effects on vocabulary memory: an activation-based model of the spacing effect*. Cognitive Science, 29(4), 559-586. DOI. Experiment/modeling; **A**, PubMed abstract. Vocabulary-specific modeling cannot directly parameterize every domain.
- R58.** Abdelrahman, G., Wang, Q., & Nunes, B. (2023). *Knowledge Tracing: A Survey*. ACM Computing Surveys, 55(11). DOI. Survey; **F**, publisher model-category and scope sections. Predictive knowledge tracing differs from causal tutoring evaluation.
- R59.** McNamara, D. S., & Kintsch, W. (1996). *Learning from texts: Effects of prior knowledge and text coherence*. Discourse Processes, 22(3), 247-288. DOI. Experiments; **A**, ERIC abstract. Coherence/knowledge interaction, not a universal sentence-length rule.
- R60.** International Baccalaureate. *Diploma Programme Subject Brief: Economics, higher level. First assessments 2022, last assessments 2029*. Official brief. Curriculum source; **O**, indexed brief. Verify the learner's assessment year and full current guide before implementing curriculum requirements.
- R61.** International Baccalaureate. *Business management guide*. Official guide. Curriculum source; **O**, indexed command-term excerpt; direct full-document fetch failed. No claim of a complete current-guide audit is made.
- R62.** Hanham, J., Leahy, W., & Sweller, J. (2017). *Cognitive Load Theory, Element Interactivity, and the Testing and Reverse Testing Effects*. Applied Cognitive Psychology. DOI. Six experiments; **A**, publisher/institutional abstract. Immediate complex-text results may differ after delay; not a general rejection of retrieval.

15. Final triage: SHIP IN V0 / TEST / REJECT

“SHIP IN V0” means scientifically defensible design requirements for a future implementation. It does not authorize building or publishing an app. “REJECT” rejects the stated claim or default, not all conceivable uses of a related technique.

Candidate	Decision	Reason / acceptance condition
Validated sources, worked solutions, and assessment keys	SHIP IN V0	Instruction cannot compensate for incorrect content
Explicit target, prerequisites, task/cue/delay conditions	SHIP IN V0	Makes learning claims assessable
Independent first attempts and preserved assistance history	SHIP IN V0	Prevents coached answers from becoming false competence evidence
Worked examples for genuinely novice complex tasks	SHIP IN V0	Strong mathematics evidence; adapt to prior knowledge
Appropriate retrieval with correction	SHIP IN V0	Strong broad evidence; retain domain/comparator boundaries
Practice distributed across sessions	SHIP IN V0	Supported broadly and in mathematics; no fixed optimum schedule
Informative feedback and fresh repair attempts	SHIP IN V0	Corrects gaps and checks whether repair can be used
Separate recall, explanation, discrimination, procedure, and transfer records	SHIP IN V0	A single number would erase outcome differences
Delayed tutor-free checks	SHIP IN V0	Directly tests the intended durability and independence
Visible equations/code/diagrams with relevant labels	SHIP IN V0	Precision and persistent structure are necessary for these tasks
Explicit programming execution model and debugging practice	SHIP IN V0	Supported design rationale and domain evidence; exact sequence remains unproven
Concise SLATE wording, stable labels, one main question per turn	TEST	Compare equal-content alternatives on time and delayed competence
2-4 sentences or 20-40 second voice chunks	TEST	Interface defaults only; revise by task and observed outcomes
Two fresh successes as a fading/criterion trigger	TEST	Practical starting rule; inadequate as high-certainty mastery proof
Targeted self-explanation prompts	TEST	Broad support with important mathematics boundary; measure added time

Candidate	Decision	Reason / acceptance condition
Exact SEE/TRACE/EXPLAIN/MODIFY/COMPLETE/GENERATE order	TEST	Plausible synthesis; allow bypass and early tiny generation
Parsons/completion bridge	TEST	Useful evidence but uncertain superiority and transfer in this setting
Interleaving onset and mixture	TEST	Depends on categories, prior knowledge, and discrimination demand
Productive-failure modules	TEST	Require suitable tasks, prerequisites, and explicit consolidation
Concrete-to-abstract fading and analogy choice	TEST	Mapping quality and prior knowledge determine usefulness
Successive relearning for explanations/procedures	TEST	Stronger direct support for learnable knowledge than arbitrary skill domains
Concept maps and prerequisite-graph optimization	TEST	Keep only if relations/learning improve beyond simpler supports
Personalized forgetting forecasts	TEST	Require delayed data, uncertainty, and future-probe validation
Review-time budget and utility thresholds	TEST	User/time allocation decisions, not cognitive laws
Optional gamification	TEST	Judge learning and time, not streaks or enjoyment alone
Universal 85% success target	REJECT	Model-specific result is not a general tutoring law
Fixed optimal sentence length or four-chunk capacity rule	REJECT	Unsupported mapping from laboratory constructs to instruction
Learning-style/medical/personality explanation of preferences	REJECT	Unsupported and unnecessary for the observed task
Literal brain programming / measured neural reconsolidation claims	REJECT	Metaphor and behavioral results cannot establish neural mechanisms
Recognition, confidence, engagement, or assisted accuracy as mastery	REJECT	They do not demonstrate the intended independent outcome
Endless hints or answer reveal before a feasible attempt	REJECT	Creates dependence and contaminates assessment
Large blank-page novice code generation as the default	REJECT	Unnecessary simultaneous demands; use small supported construction
Same-concept AB/BA crossover with assumed learning washout	REJECT	Prior teaching irreversibly contaminates the second condition

Candidate	Decision	Reason / acceptance condition
Deep learner model before an adequate event ledger/data set	REJECT	Adds complexity without validated predictive or educational value
Guaranteed far transfer or minimum-time learning claim	REJECT	Must be demonstrated under specified conditions, never promised from components alone

Decision-ready conclusion: the evidence supports a disciplined tutor with examples, practice, correction, spacing, and independent checks. ENCODE's differentiation must be demonstrated through reliable sequencing, lower total time, less dependence, and better delayed use. SLATE and personalization are candidate controls on that system, not established new cognitive mechanisms. The next authorized scientific step would be preparing and piloting verified assessment materials; this study does not claim that the proposed protocol has already worked for Advay.